

SPRINT BACKLOG REFINEMENT DOCUMENT

Project Title

AI-Driven Smart Energy Optimisation and Carbon Analytics Platform

Sprint backlog refinement refers to a process that involves evaluating, prioritising, and subdividing specific user stories into bite-sized tasks that can be accomplished during a single sprint cycle. This aids in ensuring that implementation, workloads, and objectives are well-considered.

In this project, the sprint backlog refinement process was applied to break down the overall user requirements into executable technical tasks in each sprint.

Committed User Stories for Sprint 1

User Story 1	As a system administrator, I want to establish a reliable database connection so that energy data can be stored and retrieved efficiently.
User Story 2	As a data analyst, I want all system layers to be integrated so that data can flow seamlessly from ingestion to analytics.
User Story 3	As a stakeholder, I want to visualise energy analytics through a dashboard so that system functionality can be validated.
User Story 4	As a developer, I want a basic UI framework so that future user interactions can be supported.

Outcome of Sprint Backlog Refinement

- Clear mapping between user stories and technical tasks was achieved.
- Responsibilities were assigned to each contributor.
- Backend integration and dashboard validation were successfully completed.
- UI-related tasks were identified for continuation in the next sprint.

Sprint Backlog Refinement – Task Breakdown

User Story ID	User Story Description	Task Description	Responsible	Status
US-1	Database connectivity for energy data	Design a database schema for energy data	Kavisha	Completed
		Establish database connection	Karan	Completed
		Validate data insertion and retrieval	Karan	Completed
US-2	Integrate system layers	Connect the data ingestion layer with storage	Karan	Completed
		Integrate the analytics layer with the backend	Kavisha	Completed
		Validate end-to-end data flow	Joint	Completed
US-3	Dashboard visualization	Design a Power BI dashboard layout	Kavisha	Completed
		Connect Power BI to the backend database	Kavisha	Completed
		Demonstrate a live dashboard with sample data	Joint	Completed
US-4	UI framework setup	Define UI structure and components	Karan	In Progress
		Implement initial UI screens	Joint	In Progress

Committed User Stories for Sprint 2

User Story 1	As a system administrator, I want a secure authentication system with role-based access control so that only authorised users can access the platform based on their assigned roles.
User Story 2	As an energy analyst, I want an anomaly detection engine that identifies over-consumption events with source attribution so that I can investigate and act on specific devices, zones, or departments.
User Story 3	As a sustainability manager, I want carbon emission analytics integrated into the dashboard so that I can monitor and report the organisation's carbon footprint in real time.
User Story 4	As a finance manager, I want energy cost calculations standardised in INR across all dashboard components so that financial reporting is consistent and reliable.
User Story 5	As a registered user, I want to manage my account profile and change my password from within the platform so that I can maintain my own account securely.

Outcome of Sprint Backlog Refinement

- Clear mapping between user stories and technical tasks was achieved for all five sprint objectives.
- Responsibilities were distributed between individual contributors and joint tasks based on module ownership.
- Authentication with JWT + bcrypt and role-based access control was successfully implemented and validated end-to-end.
- Anomaly detection engine was corrected, residual Z-score scoring implemented, and source attribution integrated.
- Carbon analytics tab delivered with CO2 estimation logic and live dashboard integration.
- INR cost standardisation completed — centralised tariff utility applied across all analytics modules.
- Account management pages (/profile, /change-password) delivered and integrated with the authentication module.

Sprint Backlog Refinement – Task Breakdown

User Story ID	User Story Description	Task Description	Responsible	Status
US-1	Secure authentication with role-based access control	Implement JWT token generation and validation in FastAPI auth module	Karan	Completed
		Integrate bcrypt password hashing for secure credential storage	Karan	Completed
		Define and enforce role hierarchy — Admin, User, Viewer — with tab-level access control on frontend	Kavisha	Completed
		Validate end-to-end authentication flow: login → token issue → protected route access	Joint	Completed
US-2	Anomaly detection with source attribution	Implement residual Z-score anomaly scoring: $\text{anomaly_score} = (\text{actual_kwh} - \text{baseline_kwh}) / \text{stddev}$	Karan	Completed
		Attach source attribution metadata — source_device, source_zone, source_department — to every anomaly record	Karan	Completed
		Build dual KPI metrics: Net Anomalies (scope-aware aggregate) and Any-Device Anomaly Days (device-level flag)	Kavisha	Completed
		Fix baseline propagation bug in backend query aggregation and validate corrected anomaly outputs	Joint	Completed
US-3	Carbon emission analytics on dashboard	Implement CO2 estimation logic using emission factors per energy source (Scope 1, 2, 3)	Karan	Completed
		Integrate carbon analytics tab into the Next.js dashboard with trend visualisations	Kavisha	Completed
		Connect carbon_emissions table data to frontend charts via dashboard API endpoints	Joint	Completed
US-4	Standardised INR cost calculations	Create centralised tariff utility (utils/tariff.py): $\text{cost_inr} = \text{total_kwh} \times 8.2$	Kavisha	Completed
		Refactor all analytics modules (anomaly, forecast, carbon, KPI) to consume the tariff utility	Karan	Completed

User Story ID	User Story Description	Task Description	Responsible	Status
		Update dashboard cost KPI cards to display values in INR consistently	Kavisha	Completed
US-5	User profile and password management	Develop profile update API endpoint (PATCH /auth/profile) with validation	Kavisha	Completed
		Develop password change API endpoint (POST /auth/change-password) with bcrypt re-hashing	Karan	Completed
		Build and integrate /profile and /change-password frontend pages with auth dependencies	Joint	Completed